Decision and Risk Analysis

Lecture 10: **Why Hierarchical Models?**

Complete pooling, no pooling, and partial pooling

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Alexander Abuabara



Learning goals

- Recognize grouped or repeated-measures data.
- Explain **complete pooling** and **no pooling**.
- Identify limitations of both extremes.
- Explain why **partial pooling** can improve inference and prediction.
- Distinguish within-group from between-group variation.

Reading: *Bayes Rules!*, Chapter 15:
`https://www.bayesrulesbook.com/chapter-15`

A motivating question

Suppose we measure the running time of the *same people* in several races.

- Do all runners have the same typical time?
- Does time change with age at the same rate for everyone?
- Is one runner's 2024 time informative about their 2025 time?

Key idea

Observations can be related because they belong to the same person, school, laboratory, or other group.

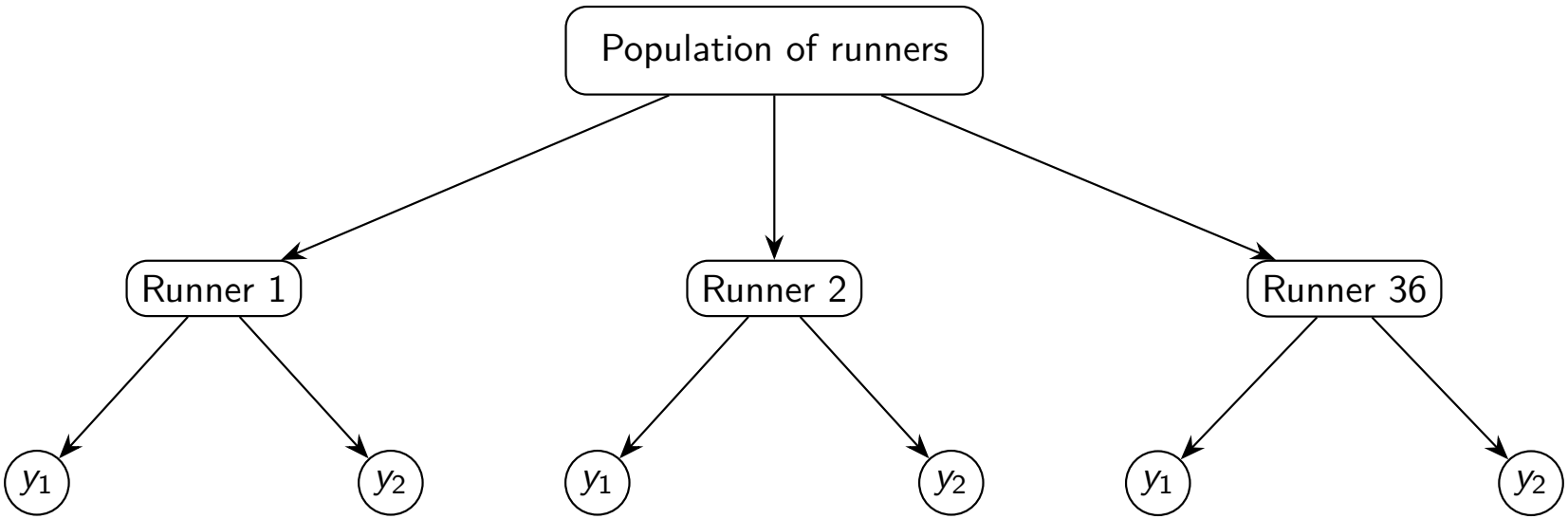
The Cherry Blossom race data

- Annual 10-mile race in Washington, D.C.
- **252 race results** from **36 runners**, each observed in multiple years.
- Runners were in their 50s or 60s.
- Y: net finish time (minutes); X: age (years).
- j indexes runners; i indexes race results within runner.

Question

How does a runner's finishing time change as they age?

What makes these data hierarchical?



- **Between-runner** variation: typical times differ across runners.
- **Within-runner** variation: times change across races for one runner.

One runner's two observations

For runner 1, the book reports:

Runner	Age	Time (min)
1	53	84.0
1	54	74.3

- Two results for the same runner need not be independent *marginally*.
- People have stable differences in ability and fitness.
- Two points alone do not establish a long-term age trend.

Would you treat these measurements as two unrelated people? Why not?

Approach 1: Complete pooling

Strategy

Put all 252 results together and fit **one regression line** for everyone.

$$Y_i \mid \beta_0, \beta_1, \sigma \overset{\text{ind}}{\sim} N(\beta_0 + \beta_1 X_i, \sigma^2).$$

- One intercept β_0 and one age slope β_1.
- Runner identity does not appear in the model.
- Assumes conditional independence of all observations given parameters.

Complete pooling: what do we estimate?

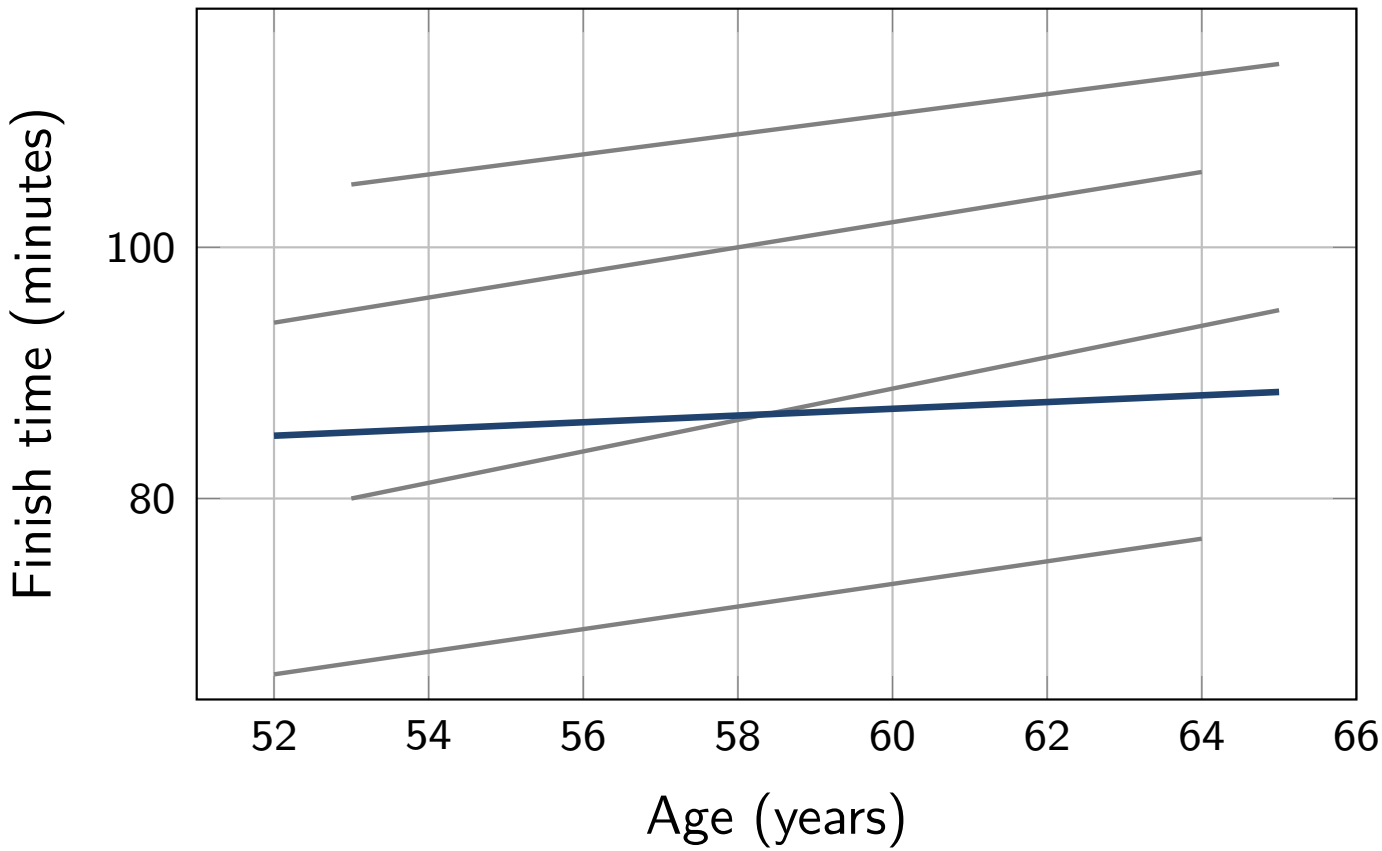
Result reported in Chapter 15

Posterior median age slope: $\hat{\beta}_1 = 0.268$ minutes/year.

$$80\% \text{ credible interval for } \beta_1 : \quad [-0.302,\ 0.842].$$

- The pooled relationship looks weak and uncertain.
- The interval includes zero; the posterior does not establish a clear positive effect at this interval level.
- But this pooled model may miss *within-person* aging patterns.

Illustration: many individual trends, one pooled trend



Schematic trends, **not** a reconstruction of the dataset.

Gray: individuals; blue: pooled.

Why complete pooling can fail

1. **Dependence:** observations from one runner share runner-specific characteristics.
2. **Heterogeneity:** different runners have different starting levels and potentially different age slopes.

Important distinction

A weak *cross-sectional* age trend does not imply that a person's time barely changes *as that person ages*.

- Ignoring the group structure can distort uncertainty and conclusions.
- This is a modeling issue, not a failure of Bayesian statistics.

Approach 2: No pooling

Strategy

Fit a **separate regression** to every runner.

$$Y_{ij} \mid \beta_{0j}, \beta_{1j}, \sigma \sim N(\beta_{0j} + \beta_{1j} X_{ij}, \sigma^2).$$

- Runner j has their own intercept β_{0j}.
- Runner j has their own slope β_{1j}.
- No group-level structure connects the regression coefficients across runners.

No pooling: runner 1 thought experiment

- Runner 1 has only about five observed races (book).
- A separately fitted line suggests that this runner becomes faster with age.
- Extrapolating that line can produce an implausibly fast prediction at age 62.

Think–pair–share

Should we trust one runner's downward-sloping line, or use information from the other 35 runners as well?

Why no pooling can fail

1. **Small groups are noisy:** a slope based on five results may be unstable.
2. **No borrowing of information:** trends from other runners cannot help estimate this runner's trend.
3. **New groups:** there is no fitted group-specific line for a runner never observed before.

Tradeoff

More flexibility is not automatically better prediction.

Approach 3: Partial pooling

Strategy

Give each runner their own trend, **but connect those trends through a shared population distribution**.

- Runners may differ in average time and rate of aging.
- Evidence from all runners informs population-level patterns.
- Group-specific estimates are *partly pulled* toward a common pattern.
- This regularization is often called **shrinkage**.

A preview of a hierarchical model

Observation level:

$$Y_{ij} \mid \beta_{0j}, \beta_{1j}, \sigma \sim N(\beta_{0j} + \beta_{1j} X_{ij}, \sigma^2).$$

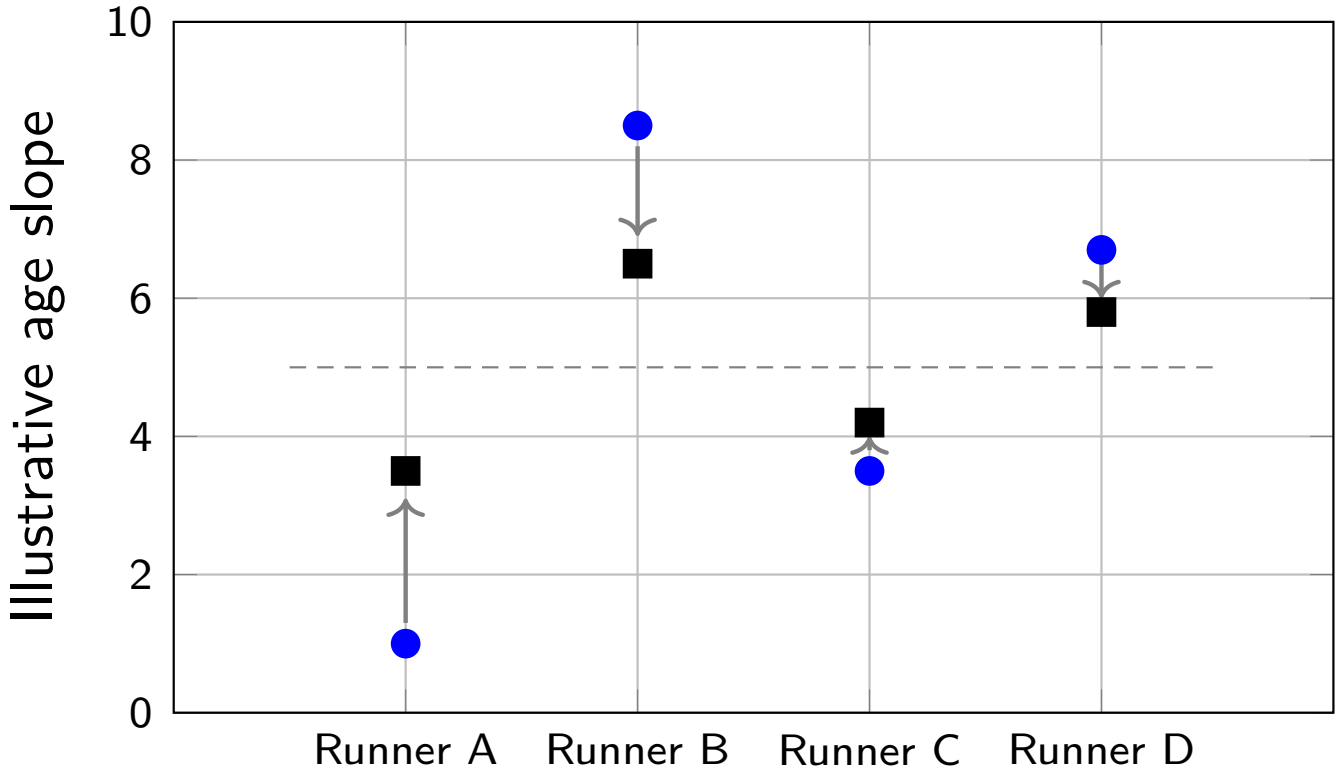
Runner level: (illustrative specification)

$$\begin{pmatrix} \beta_{0j} \\ \beta_{1j} \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_0 \\ \mu_1 \end{pmatrix}, \Sigma \right).$$

- (μ_0, μ_1): population-average coefficients.
- Σ: between-runner variation and association in coefficients.
- Priors on population parameters complete the Bayesian model.

Preview: Chapter 15 motivates this model; later chapters develop it.

Visual intuition: shrinkage



Schematic illustration only:
- Blue dots: no-pooling estimates;
- Black squares: partial-pooling estimates;
- Dashed: shared population tendency.

How much pooling?

- **Little data for a runner**: individual estimate tends to be more uncertain and often shrinks more.
- **Lots of precise data for a runner**: its own data may dominate, so shrinkage is smaller.
- **Runners are similar**: pooling is often stronger.
- **Runners are very different**: pooling may be weaker.

Intuition

Partial pooling learns the strength of sharing from the data and model rather than forcing everyone to be identical.

Comparison: all three approaches

	Runner-specific?	Shares information across runners?
Complete pooling	No	Yes; forces one common line
No pooling	Yes	No
Partial pooling	Yes	Yes; estimates related through population model

In one-sentence ...

Complete pooling ignores differences;
no pooling ignores similarities;
partial pooling models both.

Predicting a brand-new runner

Imagine a runner not among the original 36.

- **Complete pooling:** use the one universal line; may miss aging patterns.
- **No pooling:** no estimated line exists for this new runner.
- **Partial pooling:** use the learned population distribution to describe a new runner's possible intercept and slope.

Question:

Why should a prediction for a new runner include *between-runner* uncertainty?

Example beyond running: schools

We collect test scores from several students within each school.

- **Groups:** schools.
- **Observations:** students within schools.
- **Within-school variability:** students at one school have different scores.
- **Between-school variability:** schools differ in typical scores.

Question

Suppose one school has only three sampled students. How might information from other schools improve estimation?

R: explore grouping!

```
library(bayesrules)
library(dplyr)
library(ggplot2)

data(cherry_blossom_sample)
running <- cherry_blossom_sample |>
  select(runner, age, net)

# See how finish times vary by runner
ggplot(running, aes(runner, net)) +
  geom_boxplot() +
  labs(x = "Runner", y = "Finish time (min)")

# Compare within-runner patterns
ggplot(running, aes(age, net, group = runner)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE)
```

Check

For each situation, choose the modeling concern.

1. A model assumes every school has the same mean score.
2. A model estimates a separate slope from just two observations per person.
3. A model estimates separate slopes but learns a shared population trend.

Choices: complete pooling, no pooling, partial pooling.

Answers and takeaways

1. **Complete pooling**: may hide differences among schools.
2. **No pooling**: group estimates may be unstable with little data.
3. **Partial pooling**: allows variation while sharing information.

Remember

Grouped data call for models that respect the groups. Partial pooling provides a principled compromise between one-size-fits-all and completely separate analyses.

Possible questions/questions to practice

1. Give an example of data with observations nested within groups.
2. Define *within-group* and *between-group* variation in your example.
3. Explain, in two sentences, why partial pooling may be preferable to both extremes.